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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: 26-IT-56

1. Topic (12 words max)

Smart Emergency Traffic Light Control System

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	-
Year	-

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5. Brief description of the research problem including references (300 words max) – references not included in word count.

Rapid urbanization and ever-increasing vehicle ownership have considerably worsened traffic congestion in urban areas, especially in developing countries like Sri Lanka. Traffic signal systems at most existing intersections still function based on fixed-time or semi-automated control policies, which cannot adapt to real-time variations of traffic flow and grant priority to emergency vehicles such as ambulances, fire brigades, and police vehicles [1]. As a result, emergency vehicles usually suffer from severe delays when passing through signalized intersections, even in critical life-threatening situations.

The main users affected in this issue are ambulance service providers, patients in need of urgent attention for medical emergencies, Traffic Police officials, and general commuters. From the focus of user personas, ambulance drivers and paramedics are affected directly as they have issues crossing junctions even as they use the sound of alarms to maneuver through congested junctions. Patients are affected indirectly as a consequence of delayed responses.

The effects of the problem are wide-ranging. Social effects include increased mortality rates and loss of trust by the community against emergency services if there are delays in responding to emergency services [2]. The technical problem arising from the lack of real-time traffic density measurement and the tracking of emergency vehicles means that there are inefficiencies in the control systems for traffic signals. Economic effects include fuel wastages, loss of productivity, and high operation costs due to traffic congestion [3]. There are ethical concerns about safety and justice if emergency vehicles are denied priority passage.

References

[1] H. Sharma, R. Patel, and M. Shah, "Smart Traffic Light Control System Using IoT and Artificial Intelligence," IEEE Access, vol. 9, pp. 145230–145242, 2021.

[2] World Health Organization, Global Status Report on Road Safety, Geneva, Switzerland, 2023.

[3] S. K. Singh and R. S. Tomar, "Emergency Vehicle Priority Systems at Signalized Intersections: A Review," International Journal of Intelligent Transportation Systems, vol. 15, no. 3, pp. 201–210, 2020.

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6. Brief description of Existing Research and Systems (300 words max)

Traffic management methods and systems currently in use in urban areas mainly consist of fixed-time and semiautomated traffic signal control systems. The traffic signal control system works according to fixed signal times and is not adaptive regarding real-time conditions on the roads. In addition, although manual traffic control and monitoring systems and centers have been incorporated in traffic management in some urban cities, they remain inefficient in case of an emergency.

For overcoming the issue related to delays caused by emergency vehicles, some commercial as well as prototype systems have been presented, including the use of sirens and RFID, but the systems based on sirens are unreliable in noisy conditions, and systems based on RFIDs are range-limited and additional infrastructure is also required [2].

In parallel, academic research has explored intelligent traffic signal control using Internet of Things (IoT), Artificial Intelligence (AI), and machine learning techniques, including camera-based traffic density estimation through computer vision to dynamically adjust signal timings [3] and GPS-based emergency vehicle tracking to provide priority at signalized intersections [1].

Despite these advancements, many existing research solutions consider traffic density estimation and emergency vehicle tracking as independent components rather than integrating both into a unified decision-making framework. Furthermore, most studies focus on handling a single emergency vehicle approaching an intersection, while limited attention has been given to complex real-world scenarios such as multiple emergency vehicles arriving simultaneously at four-way junctions.

The key gaps identified in existing systems and research include the absence of integrated traffic density and emergency vehicle data analysis, limited support for managing multiple emergency vehicles, and poor adaptability to highly congested traffic conditions commonly observed in developing countries. These limitations highlight the need for a comprehensive, intelligent, and scalable smart traffic light system tailored to real-world urban environments.

References

[1] H. Sharma et al., "Smart Traffic Light Control System Using IoT and Artificial Intelligence," IEEE Access, 2021.

[2] S. K. Singh and R. S. Tomar, "Emergency Vehicle Priority Systems at Signalized Intersections," International Journal of Intelligent Transportation Systems, 2020.

[3] M. Li et al., "Vision-Based Traffic Density Estimation," IEEE Transactions on Intelligent Transportation Systems, 2019.

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7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

The proposed solution is a Smart Traffic Signal Priority Management System for Emergency Vehicles, designed to intelligently manage urban traffic intersections by providing real-time signal priority to emergency vehicles such as ambulances, fire brigades, and police vehicles. The system integrates GPS-based vehicle tracking, camera-based traffic density detection, artificial intelligence, and cloud computing to overcome the limitations of conventional fixed-time and semi-actuated traffic signal systems.

As seen from the conceptual diagram, the system has three main input sources: GPS signals from the emergency vehicle, live images from the traffic cameras, and synchronization of real-time data from the cloud. The emergency vehicle sends continuous GPS location signals in the form of latitude and longitude values to calculate the real-time distance from the junction in the vehicle. At the same time, the live images from the cameras at the junction of four-way intersections are processed using the Convolutional Neural Network algorithm.

The core processing layer is implemented as a Smart Traffic Management Engine hosted on the cloud. This engine integrates data from all input sources and coordinates four key system components.

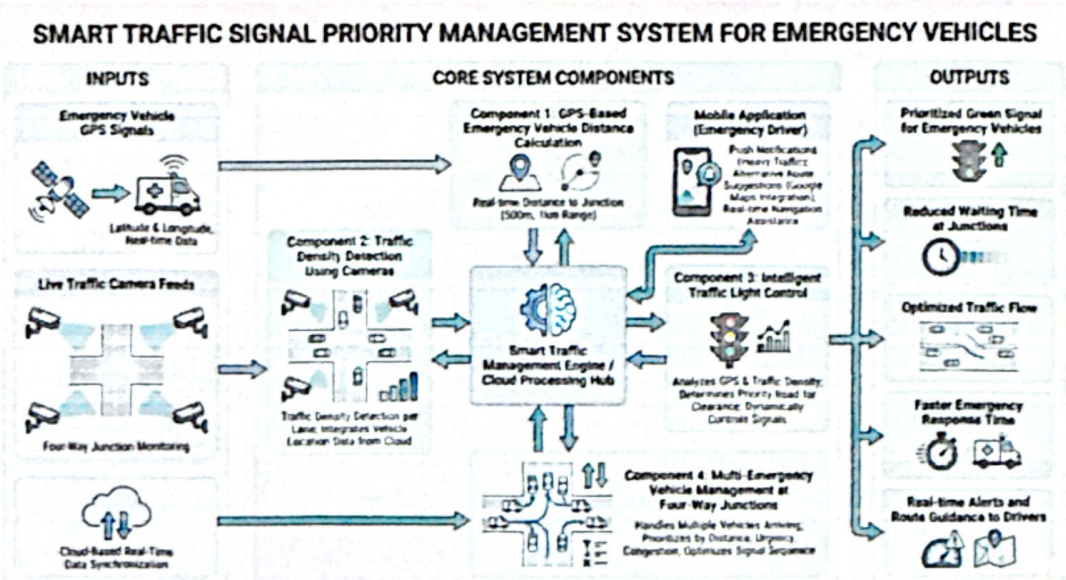
***Component 1** calculates the real-time distance of emergency vehicles from intersections using GPS data within a predefined range and A dedicated mobile application for emergency drivers acts as an auxiliary interface, providing push notifications, route suggestions, and navigation assistance*

***Component 2** performs traffic density detection using camera feeds, identifying congestion levels per lane.*

***Component 3** functions as an intelligent traffic light controller that dynamically determines the most suitable signal phase and green duration by jointly analyzing GPS and traffic density data.*

***Component 4** manages complex scenarios involving multiple emergency vehicles arriving simultaneously at four-way junctions, prioritizing vehicles based on urgency, distance, and congestion levels while optimizing overall traffic flow.*

The system outputs include prioritized green signals for emergency vehicles, reduced waiting times at junctions, optimized traffic flow for regular vehicles, faster emergency response times, and real-time alerts and route guidance. By leveraging centralized cloud processing, the proposed solution supports scalability, real-time decision-making, and coordination across multiple intersections.



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8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed Smart Traffic Light System with Emergency Vehicle Priority requires specialized domain knowledge across multiple technical and interdisciplinary areas, including traffic engineering, Internet of Things (IoT), artificial intelligence (AI), Convolutional Neural Networks (CNN), image and video processing, GPS-based vehicle tracking, and cloud computing. These domains are essential for designing an intelligent, adaptive traffic control system capable of responding to real-time traffic conditions while prioritizing emergency vehicles efficiently. The project team can acquire the required expertise through a combination of online learning platforms, academic research literature, and hands-on experimentation using open-source tools and development environments.

Technical knowledge related to AI, CNNs, and computer vision can be developed through structured online courses offered by platforms such as Coursera, edX, and Udemy, as well as through open educational resources and tutorials. CNNs play a critical role in this project, particularly in camera-based traffic analysis. CNN-based models are highly effective in extracting visual features from images and video streams, enabling accurate vehicle detection, traffic density estimation, and emergency vehicle identification at traffic junctions. By using Python-based libraries such as OpenCV, TensorFlow, and PyTorch, the project team can design, train, and evaluate CNN models for real-time traffic monitoring and decision-making. Practical implementation and experimentation further enhance the team's understanding of model performance, accuracy, and computational efficiency.

The type of data that will be used for this project includes images of traffic, real-time video images, the GPS position of emergency vehicles, and traffic density information. Real-time images and videos will be obtained from traffic cameras mounted at crossroads. For the purpose of development and simulation of the project, datasets from public sources like Kaggle may be used to allow the development of CNNs without necessarily relying on real-time images and videos. The GPS position of emergency vehicles will be obtained through the integration of GPS modules in the vehicles or the use of Google Maps APIs for the purpose of simulating the project.

All collected data can be transmitted to cloud platforms such as Firebase, Google Cloud, or Amazon Web Services (AWS) for centralized storage, real-time processing, and system coordination. Cloud computing facilitates scalability, remote access, and efficient integration of multiple data sources, making it suitable for smart city applications involving multiple intersections and continuous data flow.

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9. Objectives and Novelty

Main Objective *To design and implement a smart traffic light system that dynamically prioritizes emergency vehicles by integrating real-time traffic density detection, GPS-based emergency vehicle tracking, cloud-based data processing, and intelligent traffic signal control to minimize delays and improve emergency response efficiency.*			
Member Name with Registration No	Sub Objective	Tasks	Novelty
SURAWEERA R.B.J.H IT22298676	***To Implement GPS-based Emergency Vehicle Distance Detection with Mobile App Integration.***	*- Collect real-time GPS data from emergency vehicles* *- Calculate distance and estimated arrival time of emergency vehicles to junctions* *- Transmit GPS and distance data securely to the cloud* *- Develop a mobile application for emergency vehicles to activate emergency mode* *- Send real-time smart notifications to emergency vehicle drivers regarding traffic status (Heavy Traffic, Lane clear)*	*Integration of real-time GPS tracking with traffic control decisions, providing accurate emergency vehicle prioritization. Use of a mobile app to digitally manage emergency vehicles and receive smart traffic notifications for intelligent traffic signal control*

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JAYAMPATHI T.A.D.T IT22294234	***To implement an algorithm to calculate traffic density at a four-way junction using a camera.***	*- Monitoring real-time road images/videos using cameras* *- Detect vehicles and calculate traffic density using image processing* *- Send traffic density data to the cloud for further processing*	*Real-time traffic density estimation using camera-based vision instead of fixed-time signals or manual inputs, enabling adaptive and location-specific traffic control*
SIRIWARDANA W.L.N.S IT22070494	***To design an intelligent traffic light control system using GPS and traffic density data.***	*- Analyze traffic density and emergency vehicle distance from the cloud* *- Automatically adjust traffic light phases to create a clear path for emergency vehicles* *- Ensure smooth transition back to normal traffic flow*	*Combined decision-making using both traffic density and GPS data, which is rarely implemented together in conventional smart traffic systems*
KITHSANDU E.G.H IT22606242	***To implement multi-emergency vehicle management at four-way junctions.***	*- Detect multiple emergency vehicles approaching from different directions* *- Prioritize and sequence emergency vehicle movement safely* *- Manage four-way junction traffic using a digital traffic light system*	*Novel multi-emergency vehicle handling mechanism at complex four-way junctions, reducing conflicts and ensuring safe and efficient passage for multiple emergency vehicles*

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Ms.	Shashika	Lokuliyana	
Co-Supervisor	Mr.	Ashvinda	Idlomalgod	
External Supervisor	–	–	–	–
Summary of external supervisor's (if any) experience and expertise				

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